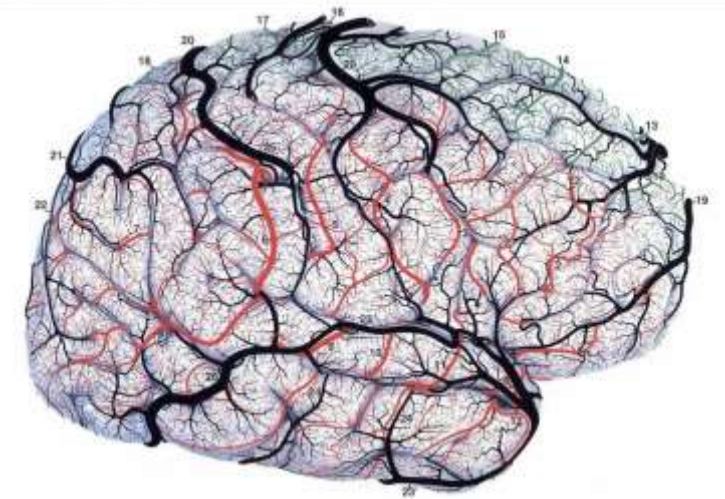
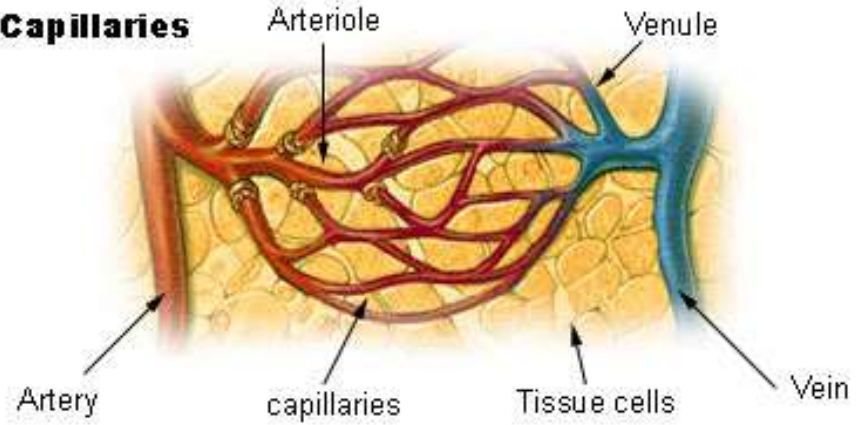


Capillaries

Biofluidynamics

Exercise 7

Blood regulation competition

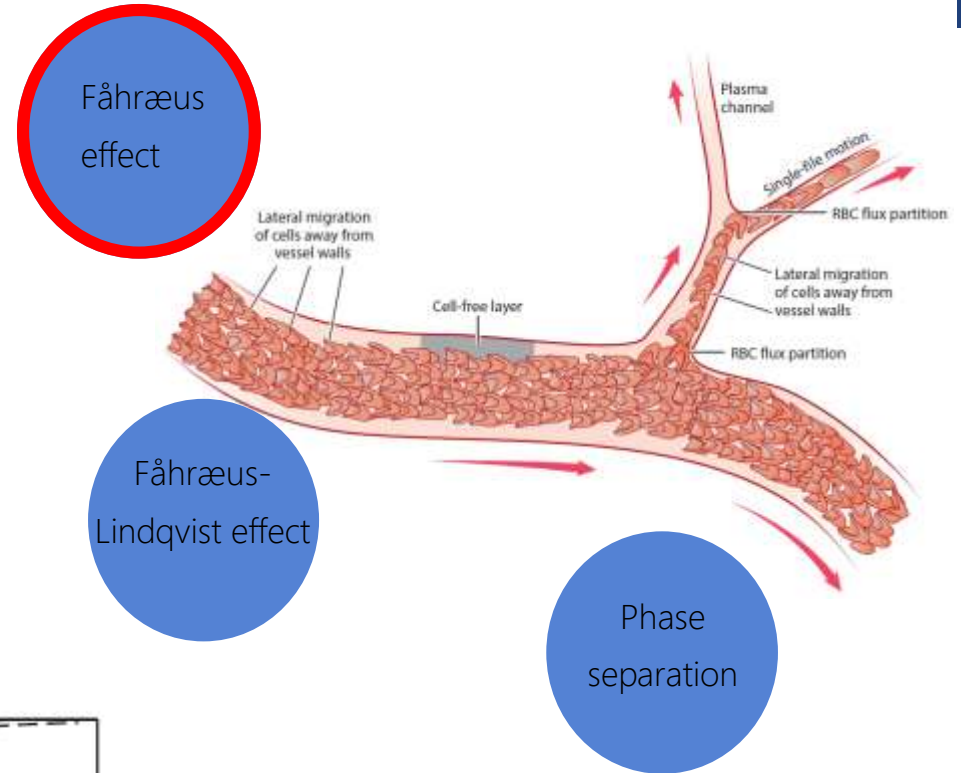
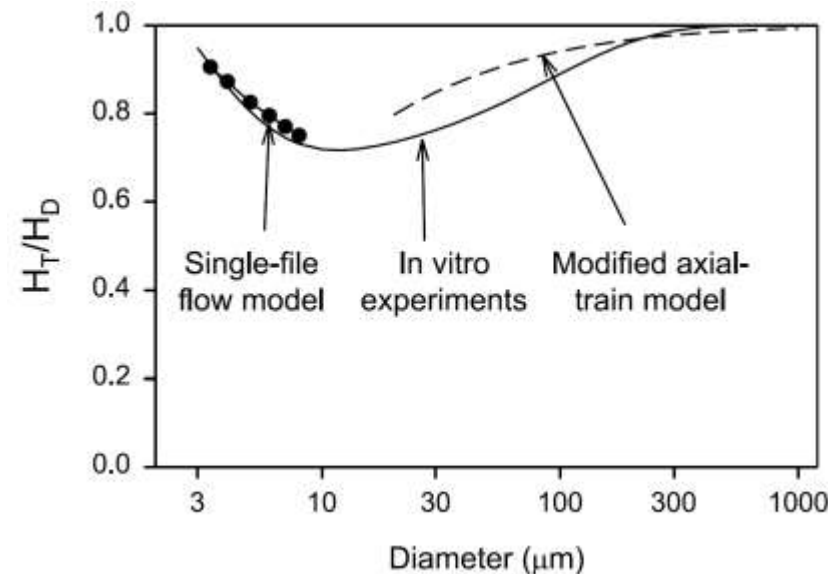
Schedule

#	Date	Lecture	Exercise
1	20. February 2026	Introduction to Biofluidynamics	Exercise #1: Dynamical scales of physiological flows
2	27. February 2026	Blood Rheology	Exercise #2: Axial train model
3	6. March 2026	Flow in Large Blood Vessels	Exercise #3: Pulsating blood flow in the femoral artery of a dog
4	13. March 2026	Cardiovascular System	Exercise #4: Arterial branchings and pulse propagation
5	20. March 2026	Respiratory flow – lung ventilation	Exercise #5: Lung ventilation
6	27. March 2026	Cardiovascular Diseases	Exercise #6: The "Windkessel" effect
		<i>Easter Break</i>	
7	17. April 2026	Microcirculation & RBC Transport	Exercise #7: Blood regulation competition 2026
8	24. April 2026	Balance sense	Exercise #8: Operating range of semicircular canals
		<i>1st May</i>	
9	8. May 2026	Respiratory flow – oxygen transport & exchange	Exercise #9: Modeling Diffusion in the Pulmonary Acinus
		<i>Ascension Day</i>	
10	22. May 2026	Cochlear mechanics	Exercise #10: Frequency-place map of the cochlea
11	29. May 2026	Repetitorium	"ETH Blood Regulator Award 2026"

Fåhræus effect

The **Fåhræus effect** describes the phenomenon that the **haematocrit of blood flowing in a long tube** (tube haematocrit H_T) is **smaller** than the **haematocrit of the discharged blood in a reservoir** (discharge haematocrit H_D).

This phenomenon is tightly connected to the fact that the average velocity of the RBCs in small vessels is higher than the mean velocity of the blood.



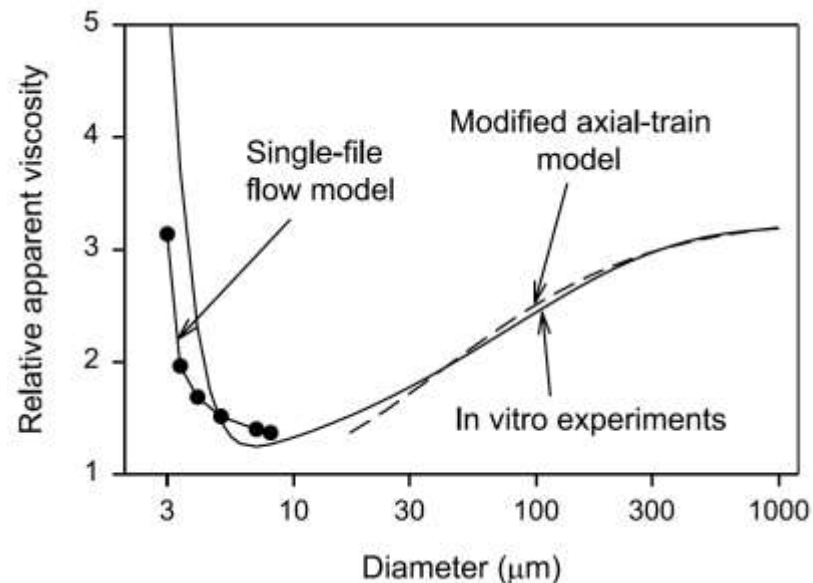
Fåhræus-Lindqvist effect

The **Fåhræus-Lindqvist effect** describe the phenomenon that the **apparent viscosity** μ_a **decreases with decreasing diameter** D when blood flows in long narrow vessels.

It reaches a minimum for $D=5-7\mu\text{m}$ which corresponds to the diameter of capillary blood vessels.

If the diameter of the capillary is smaller than $5\mu\text{m}$ there is the **reverse Fåhræus-Lindqvist effect** and the μ_a increases again.

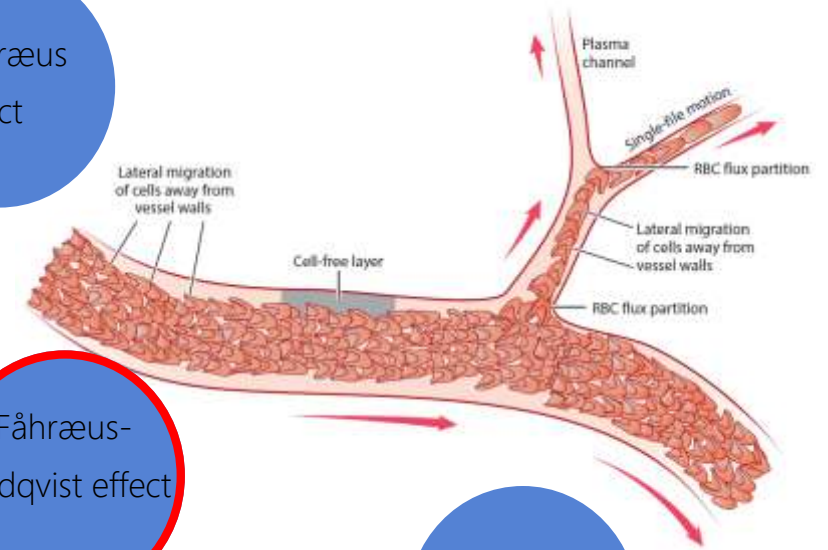
$$\mu_a = -\frac{\Delta p \pi D^4}{128 L Q}$$



Fåhræus effect

Fåhræus-Lindqvist effect

Phase separation



Phase separation

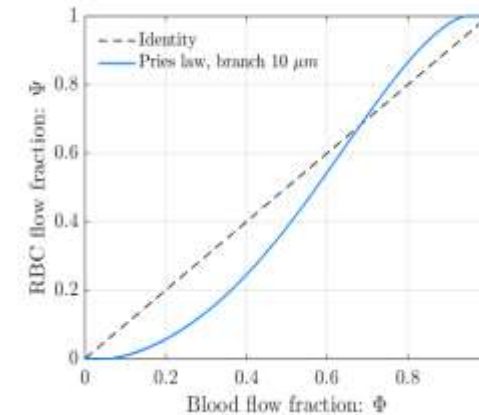
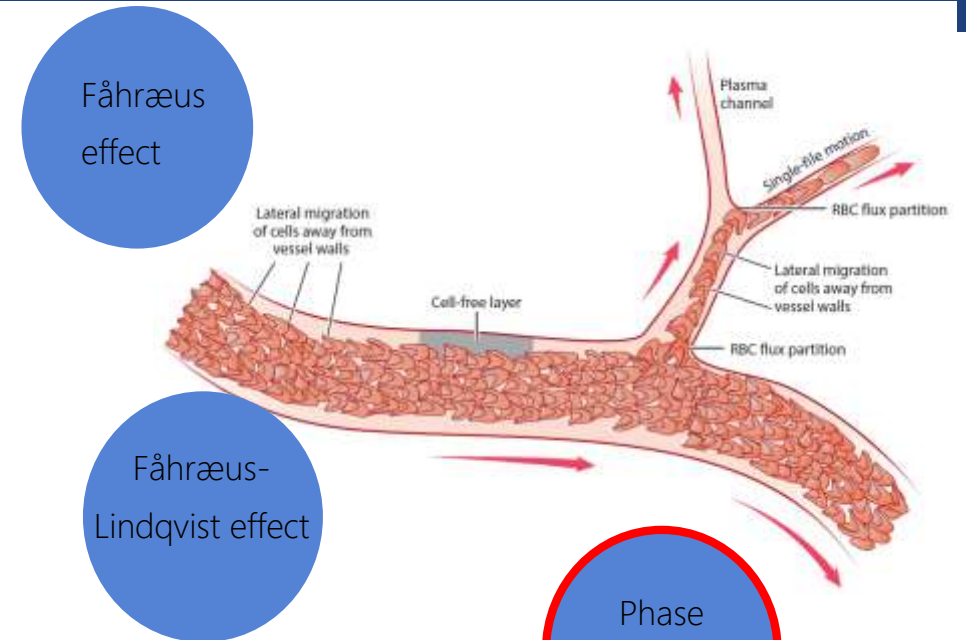
The plasma skimming mechanism is related to the axial concentration of RBC in small blood vessels. A small **daughter vessel** which obtains **fluid directly from the cell-depleted region** close to the wall will have a much **lower haematocrit** than the other daughter vessel.

The **red blood cell screening** refers to a mechanism where RBC (mainly due to their own inertia and dynamics) **deviate from the fluid streamlines** leading to an **uneven distribution** of the RBC to the daughter vessels.

Zweifach-Fung effect:

Red blood cells are attracted to the daughter vessel with the higher flow rate.

RBC flux fraction increases nonlinearly with the flow fraction entering the branch. For the smaller vessels there is a critical flow fraction below which no RBCs enter the daughter vessel.



Secomb, *Ann. Rev. Fluid Mech.*, 49, 443-461 (2017)

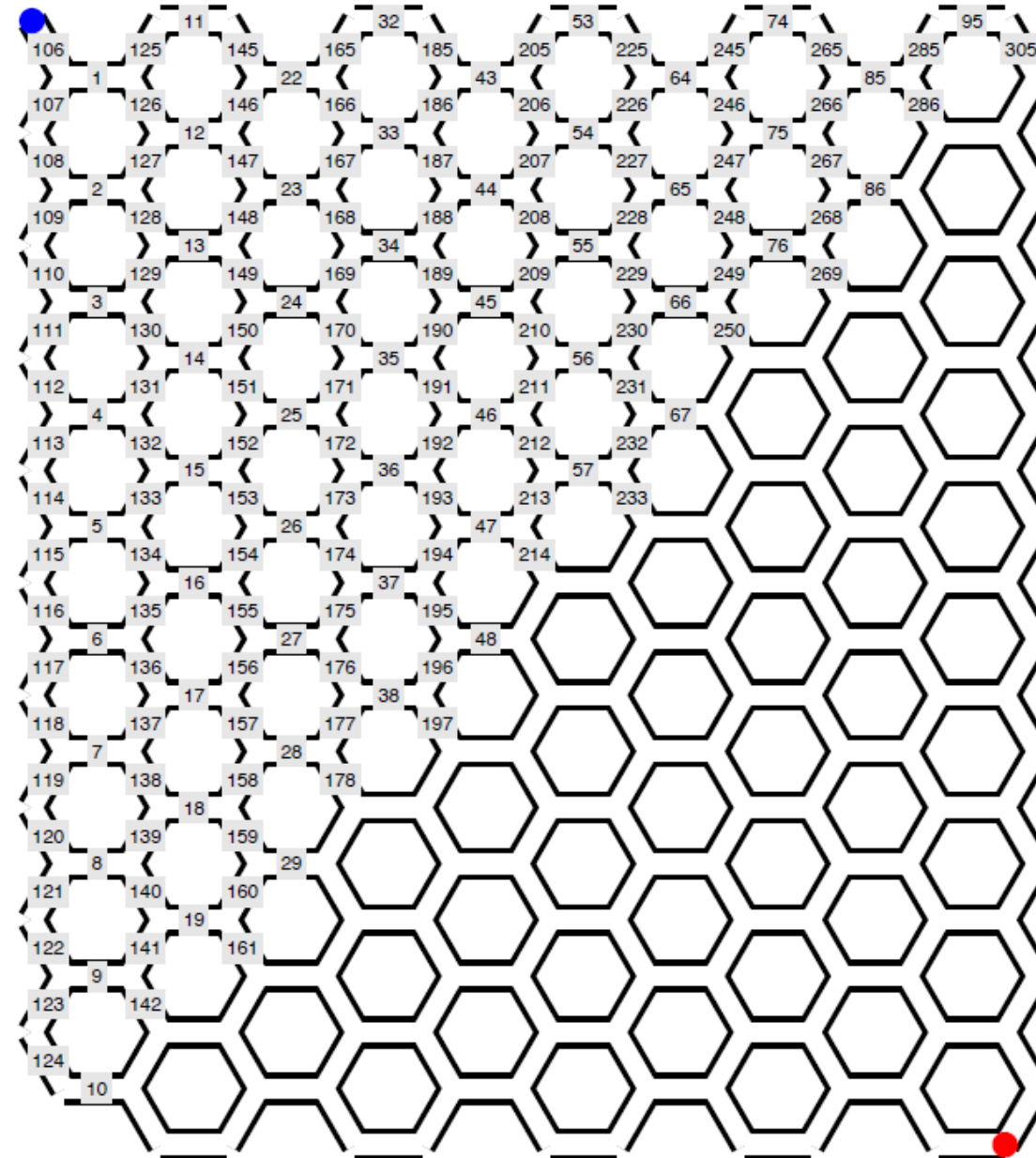
Exercise 7: Competition

This is a **mandatory** exercise that you need to pass to access the final exam in summer.

The Matlab file `competition.m` computes the red blood cell (RBC) dynamics in an idealized hexagonal capillary network. The average hematocrit of the whole network is kept constant ($\hat{H}_T = 0.25$) during the simulation, i.e. every RBC which leaves the network at the bottom-right will be immediately replaced by a new incoming RBC at the top-left. When RBCs reach a bifurcation, they will enter the capillary with the larger bulk flow velocity according to Yen et al. [1]. We further account for the Fåhræus effect by following formulae from Pries et al. [2].

Before starting the exercise:

- Run the simulation without making any changes (for this time with images).
- Observe the flow and hematocrit distribution that is plotted at the end of the simulation.
- What do you notice? Is the final distribution of RBC homogeneous?



The original network configuration is homogeneous, i.e., all vessels have equal length and radius. In this homogeneous configuration, the hematocrit distribution becomes inhomogeneous (after a transient phase). The goal of this assignment is to modify the network (by changing the vessel radii) such that the hematocrit distribution becomes more homogeneous. We will quantify the homogeneity of the hematocrit distribution by counting all the vessels that feature a mean hematocrit of $\overline{H_T} = 0.25 \pm 0.1$ (averaged from 10s to 20s of the simulation, *bin count*) and calculate the percentage to the total number of vessels. The goal is to increase this percentage from the 12,4% where no changes are applied. To receive a passing grade you need to achieve at least 16% of vessels with $\overline{H_T} = 0.25 \pm 0.1$. The one who finds the **best** solution will also be **awarded** by the celestial commission for blood flow regulation. These are the rules:

- You can only change the vessel radii in the top left half of the capillary network (see Figure 1).
- You can only dilate or constrict the diameters in these capillaries by 40%. The number of changes however is not limited.
- The results must be repeatable.

The submission process is done over Moodle under the Quizzes section. You need to upload the following **six** files with your name:

- name_surname_Setting.mat
- name_surname_Results.mat
- name_surname_Standard_Deviation.png
- name_surname_Flow.png
- name_surname_Bin_Count.png

Those files are automatically saved in the Figure folder where you have your `competition.m` script. The easiest way to name them correctly is to run the routine `competition.m`, insert 'name surname' in the test name input dialogue. The deadline is by **May 22, 2026 at 23:59.**

The Matlab file `competition.m` calls the following Matlab functions and data:

```
create\_network.m,  
select\_capillaries.m,  
compute\_network.m,  
randcon.mat,  
randleng.m,  
bifurcation\_time.m,  
plot\_network.m,  
plot\_time\_averaged\_flow.m,  
compute\_relative\_effective\_viscosity.m,  
compute\_discharge\_from\_tube.m.
```

Within those functions you will find three matrices: The vertices p (Table 1), the capillaries c between those vertices (Table 2), and the red blood cells r (Table 3).

p	1	2	3	4	5	6	7
	x	y	P	nc	f	$H_{T,avg}$	f_{avg}
	coordinate	coordinate	pressure	number of connections to this point	flow rate across the node	time-averaged hematocrit	time-averaged flow rate

Table 1: Definition of the vertices p which set up the hexagonal network.

c	1	2	3	4	5	6	7	8	9	10
	p0	p1	T	n_{RBC}	u_{bulk}	L	H_T	a	H_D	vf
	start node	end node	transmissibility	number of RBCs inside this vessel	bulk flow velocity	vessel length	tube hematocrit	vessel radius	discharge hematocrit	velocity correction factor

Table 2: Definition of the capillaries c between two vertices p0 and p1.

r	1	2	3	4	5	6
	x	y	ic	s	T_{age}	T_{remain}
	coordinate	coordinate	capillary ID	position along the vessel ($s=0$: p0, $s=1.5 \cdot 10^{-5}$: p1)	age of the RBC	time until RBC reaches next bifurcation

Table 3: Definition of the red blood cells r which flow through the capillaries.

Feel free to reach me at:

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